PART 1: Wiki Page Deliverable

🛡️ AI Architecture Governance Pack: Marketing Automation & Agentic System

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- **System Scope:** Autonomously segments audiences, generates personalized multi-channel campaign copy, and executes outreach workflows using RAG over CRM and campaign history data.
- **Core Architecture Principles:**
  - **Privacy-by-Design (AP-01):** Tokenizes or redacts customer PII and verifies marketing consent before agent invocation.
  - **Bounded Autonomy & HITL (AP-02):** Enforces least-privilege tool scopes and triggers asynchronous human approval for bulk sends (>500 recipients) or promotional discounts (>15%).
  - **Tiered Intelligence & FinOps (AP-03):** Routes tasks to the smallest capable model and enforces strict token/cost budgets.
  - **Eval-Driven Deployment (AP-04):** Requires automated quality, groundedness, and safety evaluations against a golden dataset prior to production release.
- **Architecture Decision Records (ADRs):**
  - **ADR-001 (LLM Hosting):** Adopts a hybrid approach using local PII scrubbing inside the VPC combined with an internal AI Gateway to Enterprise Zero-Data-Retention (ZDR) APIs.
  - **ADR-002 (Vector Database):** Utilizes PostgreSQL with pgvector to unify relational metadata and vector search (<5M vectors) for atomic compliance with privacy erasure deletes.
  - **ADR-003 (Execution Architecture):** Implements a dual-path architecture separating real-time event queues for high-intent triggers from asynchronous batch processing for bulk campaigns.
- **ARB Governance Checklist:** Mandates automated CI and manual sign-offs across five pillars: Security (prompt injection guardrails, least-privilege RBAC), Privacy (PDPL compliance, consent checks), Evals (>95% groundedness), Cost (token and budget limits), and Human Oversight (HITL gates, emergency kill switches, and immutable audit logs)

PART 2: Draw.io Architecture & Governance Diagram

